

 exercise-1

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Lyssna

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ID: SYS-01

Description: ESP32C6 and FRDM-MCXN236 shall be able to communicate to each other via I2C.

Rationale: I2C provides a simple, reliable, two-wire interface for bidirectional communication between the microcontrollers

Acceptance Criteria: The I2C interface shall operate at 400 kHz, 7-bit addressing shall be used and the slave address shall be 0x55

ID: SYS-02

Description: On startup, ESP32C6 shall automatically attempt to connect to Internet via configured WiFi network and

obtain the current date and time from an NTP server and use the received time to initialize its internal RTC.

Rationale: To ensure that ESP32C6 maintains an accurate real date and time.

Acceptance Criteria:

- On startup, ESP32C6 shall automatically attempt to establish a connection to the configured WiFi network.
- If the connection fails, ESP32C6 shall retry to connect to the WiFi network.
- After establishing an Internet connection, ESP32C6 shall synchronize its date and time with the configured NTP server.
- The used NTP server shall be pool.ntp.org
- The configured time zone shall be CET-1CEST,M3.5.0/02:00:00,M10.5.0/03:00:00
- After time synchronization, the Internet connection shall be terminated.

ID: SYS-03

Description: On receiving a request from FRDM-MCXN236 to get the current date and time, ESP32C6 shall

read and structure its internal RTC date and time and send it to FRDM-

MCXN236

Rationale: To enable FRDM-MCXN236 to obtain the current date and time from ESP32C6 without requiring a direct connection to an NTP server.

Acceptance Criteria: The request shall be identified by the character 'S' and ESP32C6 shall return the current date and time in `struct tm` format.

ID: SYS-04

Description: The user shall be able to interact with FRDM-MCXN236 through a terminal via serial

Rationale: To provide the user a simple interface to communicate with FRDM-MCXN236

Acceptance Criteria:

- The serial communication shall be configured as 115200-8-N-1
- The user shall be able to send commands from the terminal to FRDM-MCXN236.
- FRDM-MCXN236 shall correctly receive and process the commands and provide corresponding responses through the terminal.

ID: SYS-05

Description: FRDM-MCXN236 shall display a menu allowing the user to select one of the following options:

S) Synchronize Date and Time

D) Display the Current Date and Time

Enter a command:

Rationale: To provide the user with a simple interface for synchronizing and viewing the current date and time.

Acceptance Criteria:

- The menu shall display both options S and D.
- Selecting S shall initiate date and time synchronization.
- Selecting D shall display the current date and time.
- The user shall be able to select either option through the terminal.

ID: SYS-06

Description: FRDM-MCXN236 shall read and filter user inputs received through the serial terminal.

Rationale: To ensure that only valid user commands are processed by the FRDM-MCXN236.

Acceptance Criteria:

- FRDM-MCXN236 shall accept valid commands
- Invalid inputs shall not trigger any system function.
- The valid inputs are 's', 'S', 'd', 'D', LF and CRLF.

ID: SYS-07

Description: When the user selects 's' or 'S', FRDM-MCXN236 shall send a request to the ESP32C6 via I2C to obtain the current date and time.

Rationale: To allow FRDM-MCXN236 to obtain the current date and time from ESP32C6.

Acceptance Criteria: The commands 's' and 'S' shall trigger the date and time request and FRDM-MCXN236 shall send 'S' as the I2C request to ESP32C6.

ID: SYS-08

Description: Upon receiving the current date and time from ESP32C6, FRDM-MCXN236 shall validate

the received date and time and, if valid, set its internal RTC.

Rationale: To ensure that the FRDM-MCXN236 internal RTC is set with a valid and accurate date and time.

Acceptance Criteria:

- The received date and time shall be structured as a struct tm.
- An invalid date and time shall not modify the internal RTC of FRDM-MCXN236.

ID: SYS-09

Description: FRDM-MCXN236 shall print a success message to the terminal if date and time synchronization was successfully done;

otherwise, it shall print an error message.

Rationale: To inform the user whether the date and time synchronization was successful or not.

Acceptance Criteria:

- For successful synchronization, the terminal shall display: Synchronized successfully.

- If synchronization fails, the terminal shall display: Synchronization failed.
- Each message shall be terminated with CRLF and after displaying the message, the menu shall be displayed again.

ID: SYS-10

Description: When the user selects 'd' or 'D', FRDM-MCXN236 shall read its internal RTC, format and display the current date and time in the terminal.

Rationale: To allow the user to view the current date and time of the FRDM-MCXN236 internal RTC.

Acceptance Criteria:

- The commands 'd' and 'D' shall trigger the display of the date and time.
- The date and time shall be displayed in the format YYYY-MM-DD hh:mm:ss.
- The output shall be terminated with CRLF and after displaying the date and time, the menu shall be displayed again.

ID: SYS-11

Description: When the user presses Enter, FRDM-MCXN236 shall display the menu.

Rationale: To allow the user to return to the menu to be able to select an option.

Acceptance Criteria: LF and CRLF shall be recognized as a new-line input and no system function shall be triggered.